



Technical Service Bulletin

Technical Service Bulletin: TSB110301

Released Date: 28-Oct-2016

CM850 and CM2150 (MCRS) FR, DO, Engine Control Module (ECM), and CPL Cross-References

CM850 and CM2150 (MCRS) FR, DO, Engine Control Module (ECM), and CPL Cross-References

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Cummins Inc. has released revisions to the Modular Common Rail System (MCRS) fuel system since its introduction. If an engine is upgraded from C1 to C2, C1 to C3 or C2 to C3, it requires a CPL, FR, DO, and ECM code change. The engine data plate **must** be stamped with the new CPL, FR, and DO options and Engine Modification Form 1877 **must** be completed.

Cummins Inc. recommends upgrading the engine to the C3 fuel system during rebuild events to minimize the cost and improve fuel system durability relative to fuel quality issues. Reference TSB100978 for information about the parts required to upgrade.

Use the file below to find the appropriate CM850 to CM2150 FR, DO, and ECM Code for C2 to C3 upgrades:

QSK19							
Customer	Application	CM850 (C2)			CM2150 (C3)		
		FR Option	DO Option	ECM Code	FR Option	DO Option	ECM Code
Al's Diesel	Mud Pump	4463	4736	AQ40108	4551	4900	AQ40210
Atlas Copco	RD20 Drill	4464	4639	AQ40045	4511	4732	AQ40089
Atlas Copco	Drill	4465	4640	AQ40049	4549	4796	AQ40137
Atlas Copco	Drill	4465	4726	AQ40111	4549	4877	AQ40186

QSK19							
Atlas Copco	Drill	4464	4768	AQ40116	4511	4878	AQ40187
Atlas Copco	MT 5020	4466	4946	AQ40237	4549	4957	AQ40240
Baker Corp	Centrifugal Pump	4467	4716	AQ40091	4510	4893	AQ40207
Bandit	3680/4680 Beast	4463	4718	AQ40092	4551	4805	AQ40143
Bandit	337-FCX Chipper	4466	4699	AQ40079	4556	4817	AQ40160
Belaz	7555F	4466	4721	AQ40097	4556	4856	AQ40174
BJ Services	Acid Frac	4463	4646	AQ40044	4551	4804	AQ40142
Bonny China	CE125 Excavator	4467	4739	AQ40095	4510	4894	AQ40218
Bucyrus Intl	SKF Blasthole Drill	4465	4954	AQ40244	4549	4955	AQ40241
Compression LEA	Air Compressor	4465	4723	AQ40099	4549	4899	AQ40209
Compression LEA	Booster Combo	4464	4784	AQ40122	4511	4905	AQ40215
Crown Energy	ST80080 Blender	4464	4662	AQ40066	4511	4876	AQ40185
Cummins Power	Generic	4466	4650	AQ40051	4556	4816	AQ40159
Cummins Taiwan	Emergency Pump	4468	4861	AQ40176	4564	4914	AQ40255
Diamond Z Mfg	Tub Grinder	4466	4654	AQ40055	4556	4889	AQ40203
Dopplemayr	Ski Lift	4464	4698	AQ40078	4549	4898	AQ40208
Dressta/HSW	TD40 Dozer	4473	4664	AQ40060	4567	4836	AQ40165
Dressta/HSW	TD40 Dozer	4473	4729	AQ40102	4567	4880	AQ40189
Drill Tech	Rock Drill	4472	4706	AQ40084	4566	4883	AQ40192

QSK19							
Drill Tech/Sa ndvik	D75 Drill	4464	4686	AQ40069	4511	4810	AQ40148
DW/TXS	DDW- 2000 Drill	4466	4958	AQ40246	4556	4959	AQ40245
Ensign Drilling	Pump	4466	4661	AQ40067	4556	4890	AQ40204
Euclid	R60	4466	4814	AQ40151	4556	4868	AQ40181
Euclid	EH700	4472	4795	AQ40131	4566	4884	AQ40193
Flowtron ex	Water Pump	4467	4697	AQ40077	4510	4892	AQ40206
Foremos t	DR40	4466	4792	AQ40124	4556	4857	AQ40175
Foremos t	DR24 HD	4466	4819	AQ40154	4556	4865	AQ40178
Foremos t	DR24	4501	4794	AQ40129	4565	4879	AQ40188
General Equip	NA	4466	4659	AQ40059	4556	4855	AQ40173
Harsco Track	Rail Grinder	4467	4678	AQ40068	4510	4870	AQ40183
Hokuriku	Hydraulic Loco	4466	4848	AQ40177	4556	4864	AQ40198
Hydra Rig	Well Pump	4463	4759	AQ40113	4551	4902	AQ40212
John Deere	SPFH790 0	4515	4738	AQ40101	4555	4812	AQ40150
Kawasak i	115ZA	4498	4688	AQ40071	4556	4797	AQ40138
Lamb Service	Hydraulic Pump	4463	4725	AQ40110	4551	4901	AQ40211
Lampson	Transpor t lift	4464	4802	AQ40141	4511	4907	AQ40217
Leibherr	R984C	4467	4651	AQ40052	4510	4731	AQ40088
Leitner Poma	Ski Lift	4466	4724	AQ40100	4556	4867	AQ40180
Leitner Poma	Ski Lift	4466	4740	AQ40096	4556	4869	AQ40182

QSK19							
Luigong	Wheel Loader	4472	4708	AQ40085	4566	4841	AQ40167
Manitowoc	Crane	4501	4783	AQ40121	4565	4882	AQ40191
Marine	Auxiliary	4459	4629	Aq40073	4528	4766	AQ40117
Marine	Propulsion	4462	4636	AQ40043	4530	4769	AQ40132
Marine	Propulsion	4442	4593	AQ40017	4531	4770	AQ40126
Marine	Propulsion	4489	4681	Aq40064	4532	4771	AQ40127
Marine	Propulsion	4488	4680	AQ40063	4533	4772	AQ40133
Marine	Propulsion	4450	4608	AQ40024	4534	4773	AQ40134
Marine	Propulsion	4432	4545	AQ40010	4535	4774	AQ40135
Morbark	Tub Grinder	4468	4722	Aq40098	4564	4866	AQ40179
Motion Industries	NA	4463	4947	AQ40238	4551	4948	AQ40238
Motive Power	Locomotive	4468	4695	AQ40075	4564	4891	AQ40308
Multi Power	Siesmic Drill	4467	4756	AQ40112	4510	4895	AQ40219
Novamac	Drill	4466	4843	AQ40161	4556	4897	AQ40220
NREC	Locomotive	4468	4658	AQ40058	4564	4833	AQ40169
Powermatch	Generic	4465	4702	AQ40082	4549	4808	AQ40146
Powermatch	Generic	4501	4692	AQ40072	4565	4834	AQ40163
Rolligon	Twin Pumper	4463	4645	AQ40046	4551	4798	AQ40139
Rolligon	Combo Frac	4463	4660	AQ40065	4551	4806	AQ40144
Romco	Haul Truck	4472	4719	AQ40093	4566	4909	AQ40222

QSK19							
Sandvik	D75 Drill	4468	4705	Aq40083	4564	4849	AQ40172
Schlumberger	Frac Rig	4463	4780	AQ40120	4551	4872	AQ40195
Shantui	Dozer	4547	4793	AQ40128	4553	4800	AQ10140
Soilmec	P 560 Drill	4464	4765	AQ40115	4511	4904	AQ40214
Soilmec	SC100 Drill	4464	4791	AQ40123	4511	4906	AQ40216
Southern Industries	Mud Pump	4465	4710	AQ40086	4549	4807	AQ40145
Southern Industries	Draw Works	4464	4641	AQ40048	4549	4875	AQ40184
Surefire	N2 Pump	4463	4949	AQ40242	4551	4950	AQ40239
Taylor Brothers	Crane	4501	4631	AQ40130	4565	4908	AQ40221
Terex	RH120E	4467	4652	AQ40053	4510	4733	AQ40090
Terex	TR60	4466	4655	AQ40056	4556	4815	AQ40152
Terex	TR45	4472	4665	AQ40062	4566	4835	AQ40164
Terex	3-90	4472	4663	AQ40061	4566	4840	AQ40166
Texas Recycling	Concrete Recycler	4472	4737	AQ40109	4566	4910	AQ40223
Toro	T60	4465	4642	AQ40050	4549	4809	AQ40147
Trican Well Services	Nitrogen Pump	4463	4701	AQ40081	4551	4874	AQ40196
Trican Well Services	Fluid Pumper	4463	4970	AQ40255	4551	4971	AQ40253
Truflo	Water Pump	4464	4764	AQ40114	4511	4903	AQ40213
Truflo	Water Pump	4464	4930	AQ40236	4556	4932	AQ40234
Vandel	Landfill Compactor	4501	4720	AQ40094	4565	4881	AQ40190
Xian	DR130	4464	4887	AQ40201	4511	4888	AQ40202

QSK19							
Xugong	GM600 Wheel Loader	4472	4885	AQ40199	4566	4886	AQ40200
Xugong	GM600 Dump Truck	4472	4775	AQ40119	4566	4911	AQ40224
Power Generation	G-Drive	4516	4755	Obsolete	4580	4858	AQ40276
		4516	4756	Obsolete	4580	4858	AQ40276
		4517	4757	Obsolete	4581	4859	AQ40265
		4517	4758	Obsolete	4581	4859	AQ40265
QSK38							
Customer	Application	CM850 (C2)			CM2150 (C3)		
		FR Option	DO Option	Parent ECM Code	FR Option	DO Option	Parent ECM Code
Bucyrus	RH200	6630	6586	AQ60111	6735	60063	AQ60339
Bucyrus	RH200	6630	6585	AQ60110	6735	60064	AQ60340
D.C. Otto	Water Pump	6630	6776	AQ60216	6735	60105	AQ60351
Hitachi	EX1900	6628	6525	AQ60083	6780	6851	AQ60269
Komatsu	PC3000	6630	6530	AQ60094	6735	6766	AQ60215
Komatsu	PC3000 w/Centinel	6630	6613	AQ60116	6735	6996	AQ60319
LeTourneau	L1150	6694	6670	AQ60165	6779	6850	AQ60268
Marine	Auxiliary	6579	6457	AQ60082	6757	6811	AQ60306
Marine	Auxiliary	6637	6541	AQ60106	6758	6812	AQ60307
Marine	Propulsion	6592	6469	AQ60041	6747	6795	AQ60231
Marine	Propulsion	6668	6615	AQ60122	6778	6808	AQ60263
Marine	Propulsion	6558	6360	AQ60028	6755	6809	AQ60264
Marine	Propulsion	6591	6468	AQ60042	6756	6810	AQ60265

QSK50							
Customer	Application	CM850 (C2)			CM2150 (C3)		
		FR Option	DO Option	Parent ECM Code	FR Option	DO Option	Parent ECM Code
Bayou Well Services	Hyd. Frac Pump	6709	60247	AQ60452	6888	60236	AQ60445
BJ Services Canada	Frac Rig	6717	60172	AQ60411	6890	60173	AQ60410
BJ Services Canada/ Propell	Frac Rig	6709	6937	AQ60259	6888	60104	AQ60381
BJ Services Canada/ Propell	Frac Rig	6717	60016	AQ60323	6890	60097	AQ60364
Cisco High Lift	Frac Rig	6717	6726	AQ60181	6890	60129	AQ60388
Cisco High Lift	Frac Rig	6709	6753	AQ60200	6888	60120	AQ60375
Collier Well	Frac Pumper	6710	60194	AQ60432	6889	60195	AQ60416
CS&P	Frac Rig	6709	6768	AQ60209	688	60121	AQ60376
CS&P	Frac Rig	6709	60382	AQ60461	6888	60383	AQ60476
CS&P	Frac Rig	6717	60376	AQ60460	6890	60377	AQ60475
CS&P	Frac Rig	6717	60374	AQ60457	6890	60375	AQ60456
Dragon	Frac Rig	6710	6737	AQ60194	6889	60123	AQ60367
Dragon	Frac Rig	6717	6727	AQ60182	6890	60012	AQ60361
Enerflow	Frac Rig	6717	6847	AQ60243	6890	60131	AQ60390
Enerflow	Frac Rig	6717	60113	AQ60406	6890	60135	AQ60402
Enerflow	Frac Rig	6710	6711	AQ60191	6889	60011	AQ60360
Halliburton	Frac Rig	6709	6734	AQ60192	6888	60013	AQ60362
Hitachi	EH3500	6612	6508	AQ60069	6734	6976	AQ60296
Hitachi	EX2600/ 5600	6742	6789	AQ60219	6790	6866	AQ30288

QSK50							
Hitachi	EX2500/5500	6615	6511	AQ60068	6795	6871	AQ60289
IDMS	Frac Rig	6709	60106	AQ60394	6888	60132	AQ60395
IDMS	Frac Rig	6717	60231	AQ60449	6890	60213	AQ60440
IDMS	Frac Rig	6717	60001	AQ60311	6890	60133	AQ60391
Komatsu	730E w/Centinel	6612	6510	AQ60071	6734	6928	AQ60252
Komatsu	730E	6612	6509	AQ60070	6734	6765	AQ60217
Komatsu	HD1500-7	6609	6491	AQ60056	6736	6767	AQ60220
Letourneau	L1350	6625	6524	AQ60090	6782	6853	AQ60287
Liebherr	R9400	6660	6793	AQ60223	6794	6870	AQ60421
LKAB-STT	Locomotive	6689	6540	AQ60115	6859	6980	AQ60297
Marine	Auxiliary	6638	6542	AQ60107	6762	6816	AQ60290
Marine	Auxiliary	6580	6458	AQ60038	6763	6817	AQ60291
Marine	Propulsion	6594	6474	AQ60047	6749	6797	AQ60232
Marine	Propulsion	6593	6473	AQ60046	6761	6815	AQ60273
Marine	Propulsion	6669	6616	AQ60123	6759	6813	AQ60271
Marine	Propulsion	6575	6363	AQ60029	6760	6814	AQ60272
NREC	Locomotive	6635	6539	AQ60114	6861	60215	AQ60444
Orteq	Hyd. Frac Pump	6717	60196	AQ60447	6890	60197	AQ60442
Ortowski	Frac Rig	6710	6964	AQ60280	6889	60191	AQ60441
Ortowski	Frac Rig	6710	6752	AQ60199	6889	60126	AQ60370
Petrol SJ	Frac Pump	6717	60201	AQ60448	6890	60205	AQ60443
Propell Oil	Frac Rig	6709	60056	AQ60336	6888	60100	AQ60378
Propell Oil	Frac Rig	6717	60014	AQ60321	6890	60096	AQ60363

QSK50							
Propell Oil	Frac Rig	6717	6744	AQ60195	6890	60130	AQ60389
Rolligon	Frac Rig	6709	6751	AQ60198	6888	60119	AQ60374
Rolligon	Frac Rig	6709	6935	AQ60258	6888	60103	AQ60386
Rolligon NOV	Frac Rig	6709	60153	AQ60403	6888	60154	AQ60377
Rolligon NOV	Frac Rig	6709	6874	AQ60248	6888	60122	AQ60377
Sanjel	Frac Rig	6717	60156	AQ60407	6890	60157	AQ60397
Schlumberger	Frac Rig	6709	6735	AQ60193	6888	60101	AQ60379
Surefire	Frac Rig	6709	60117	AQ60372	6888	60118	AQ60373
Total Equipment	Frac Rig	6709	60015	AQ60322	6888	60099	AQ60366
Trican	Frac Rig	6710	6750	AQ60197	6889	60125	AQ60369
Trican Custom Welding	Frac Rig	6717	6876	AQ60249	6890	60128	AQ60387
Trican/Tycrop	Frac Rig	6710	6770	AQ60211	6889	60127	AQ60371
United Equip	Frac Rig	6709	6855	AQ60262	6888	60102	AQ60380
Universal	Frac Rig	6717	60085	AQ60346	6890	60098	AQ60365
Well Bank	Frac Rig	6710	6749	AQ60196	6889	60124	AQ60368
Wirtgen	Plainer	6609	6805	AQ60222	6736	60081	AQ60338
QSK50 G-Drive							
Customer	Application	CM850 (C2)			CM2150 (C3)		
		FR Option	DO Option	Parent ECM Code	FR Option	DO Option	Parent ECM Code

QSK50 G-Drive							
Power Generati on	G-Drive	6667	6608	AQ60151	6832	6938	AQ60353
		6667	6609	AQ60152	6832	6938	AQ60353
		6621	6532	AQ60102	6833	6940	AQ60354
		6661	6611	AQ60119	6834	6941	AQ60355
		6643	6555	AQ60137	6838	6948	AQ60382
		6643	6556	AQ60138	6838	6948	AQ60382
		6726	6746	AQ60205	6878	6998	AQ60384
		6726	6747	AQ60206	6878	6998	AQ60384
		6726	60234	AQ60473	6878	6998	AQ60384
		6655	6596	AQ60141	6840	6952	AQ60383
		6655	6597	AQ60142	6840	6952	AQ60383
		6654	6593	AQ60139	6879	6999	AQ60385
		6654	6594	AQ60140	6879	6999	AQ60385
		6641	6549	AQ60143	6835	6942	AQ60356
		6641	6550	AQ60144	6835	6942	AQ60356
		6642	6552	AQ60145	6836	6944	AQ60357
		6642	6553	AQ60146	6836	6944	AQ60357
		6653	6645	AQ60153	6837	6946	AQ60358
		6653	6646	AQ60154	6837	6946	AQ60358
		6606	6528	AQ60074	6878	6998	AQ60384
		6606	6548	AQ60100	6878	6998	AQ60384
		6606	6547	AQ60103	6878	6998	AQ60384
		6730	6760	AQ60228	6839	6951	AQ60298
		6731	6762	AQ60229	6841	6955	AQ60299

QSK60							
Custome r	Applicati on	CM850 (C2)			CM2150 (C3)		
		FR Option	DO Option	Parent ECM Code	FR Option	DO Option	Parent ECM Code
Bucyrus	Excavato r	6619	6827	AQ60230	6781	6997	AQ60318
Hitachi	EH4000	6629	6526	AQ60097	6746	6975	AQ60295
Hitachi	EX3600/ 8000	6614	6521	AQ60088	6796	6872	AQ60303
Komatsu	PC8000	6613	6520	AQ60084	6792	6989	AQ60304
Komatsu	830E AC	6618	6501	AQ60067	6733	6764	AQ60218
Komatsu	830E AC	6629	6849	AQ60246	6746	6929	AQ60253

QSK60							
Komatsu	860E	6617	6513	AQ60075	6751	60003	AQ60312
Komatsu	860E	6617	6799	AQ60225	6751	6803	AQ60236
Komatsu	930E	6617	6512	AQ60077	6751	6804	AQ60237
Komatsu	930E	6724	6774	AQ60240	6791	6867	AQ60337
Komatsu	PC4000	6634	6535	AQ60099	6793	6869	AQ60305
Letourneau	L1850	6613	6518	AQ60089	6792	60002	AQ60314
Letourneau	L1850	6613	6844	AQ60242	6792	60006	AQ60316
Liebherr	PR986	6613	6519	AQ60087	6792	6868	AQ60302
Marine	Auxiliary	6652	6572	AQ60156	6772	6826	AQ60257
Marine	Auxiliary	6595	6573	AQ60157	6771	6825	AQ60256
Marine	Propulsion	6651	6571	AQ60109	6767	6821	AQ60276
Marine	Propulsion	6650	6570	AQ60108	6769	6823	AQ60278
Marine	Propulsion	6596	6499	AQ60058	6770	6824	AQ60279
Marine	Propulsion	6663	6599	AQ60120	6766	6820	AQ60275
Marine	Propulsion	6664	6600	AQ60121	6765	6819	AQ60274
Marine	Offshore Drilling	6720	6730	AQ60180	6737	6772	AQ60233
Terex	RH400	6619	6544	AQ60095	6781	6852	AQ60317
QSK60 G-Drive							
Customer	Application	CM850 (C2)			CM2150 (C3)		
		FR Option	DO Option	Parent ECM Code	FR Option	DO Option	Parent ECM Code

QSK60 G-Drive							
Power Generati on	G-Drive	6671	6617	AQ60155	6807	6900	AQ60427
		6671	6641	AQ60164	6807	6900	AQ60427
		6671	6748	AQ60207	6807	6900	AQ60427
		6644	6558	AQ60124	6808	6902	AQ60485
		6644	6559	AQ60125	6808	6902	AQ60485
		6682	6637	AQ60158	6809	6904	AQ60513
		6682	6638	AQ60159	6809	6904	AQ60513
		6683	6639	AQ60160	6810	6906	AQ60514
		6683	6640	AQ60161	6810	6906	AQ60514
		6665	6602	AQ60128	6811	6908	AQ60486
		6665	6603	AQ60129	6811	6908	AQ60486
		6665	6842	AQ60241	6811	6908	AQ60486
		6666	6605	AQ60130	6812	6910	AQ60487
		6666	6606	AQ60131	6812	6910	AQ60487
		6646	6564	AQ60147	6813	6912	AQ60515
		6646	6565	AQ60148	6813	6912	AQ60515
		6647	6567	AQ60149	6814	6914	AQ60516
		6647	6568	AQ60150	6814	6914	AQ60516
		6670	6619	AQ60132	6815	6916	AQ60488
		6670	6620	AQ60133	6815	6916	AQ60488
		6645	6561	AQ60126	6816	6918	AQ60324
		6645	6562	AQ60127	6816	6918	AQ60324
		6845	6991	AQ60308	6880	6921	AQ60428
		6671	6748	AQ60207	4580	4858	AQ40276
		6962	60182	AQ60418	4581	4859	AQ40265
		6963	60183	AQ60419	4581	4859	AQ40265
		6602	6529	AQ60073	6807	6900	AQ60427

Use the Engine CPL Cross-Reference Table below to find the appropriate engine CPL:

Engine CPL Cross-Reference Table						
Engine Model	C1 CPL	C2 CPL	C3 CPL	C1 Injector	C2 Injector	C3 Injector
QSK19 Tier III	8533	2754	3240	4928106	4964170	2867149
	8605	2755	3798			
	1470	2757	3666			

Engine CPL Cross-Reference Table						
Engine Model	C1 CPL	C2 CPL	C3 CPL	C1 Injector	C2 Injector	C3 Injector
QSK19 Tier II	563	2756	3242	4928053	4964171	2867147
	8744	2768	3455			
		2882	3437			
	1485	3232	3716			
QSK38 and QSK50		0847	3327	4928104	4964172	2867147
		1158	3328			
		2088	3464			
		2089	3465			
	1466	2758	3380			
	1925	2759	3380			
	8524	2760	3379			
	8624	2761	3548			
	1764	2762	3391			
	1492	2771	3621			
	2767	2959	3976			
		3080	3669			
		3170	3465			
		3300	3728			
		3329	3728			
		3377	3621			
QSK60		2703	3478	4928105	4964173	2867148
		2704	3476			
	8593	2763	3416			
	1663	2764	3501			
	1773	2766	3451			
	0930	2769	3578			
	8595	2770	3850			
	0931	2773	NONE			
	2726	2794	3378			
		3166	3477			

Note : If your CPL is not listed, please submit a QuickServe® Online ticket. This list does not include some C3 CPLs which are new and have no relation to older CPLs.

Document History

Date	Details
2011-11-2	Module Created
2011-11-9	QSOL Quick Fix Reason: Outfile Notes: none
2011-11-11	QSOL Quick Fix Reason: Spelling Error Notes: none
2012-12-7	Adding table of FR, DO
2013-2-1	QSOL Quick Fix Reason: Incorrect Part Number Notes: At the top of the QSK60 table it says CM2250, but it should be CM2150 like the other tables.
2013-4-8	wrong application listed
2014-4-14	none
2015-3-25	Update 'TBD' data in calibration cross reference table
2015-10-27	Updated QSK50 G-Drive with FR options 6730 and 6731. Updated cross-reference table with CPL 3377 and 3621.
2016-3-30	Added FR options 6602 and 6807 for QSK60 G-Drive. Updated cross-reference table with CPL 3578 for QSK60.
2016-10-26	Updated AQ40205 to AQ40308. Updated C3 CPL column in CPL Cross-Reference Table.

Last Modified: 28-Oct-2016
